12 Mark for Review

The function f is defined by $f(x) = 53(0.15)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

(A) 15

(B) 47

(C) 53

(D) 85

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22 Mark for Review

The function f is defined by $f(x) = ab^{\frac{x}{n}}$, where a, b, and n are constants, and b and n are integers. If $f(4) = 5$ and $f(7) = 135$, what is the value of $f(9)$?



Answer Preview:

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The function f is defined by $f(x) = b^x + c$, where b and c are positive constants. In the *xy*-plane, the graph of $y = f(x)$ passes through the point $(0, 25)$. What is the value of $f(0)$?

Enter response...

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The function f is defined by $f(x) = 55(0.19)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

A. 19

B. 45

C. 55

D. 81

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The functions f and g are defined by the given equations, where $x \geq 0$, a and b are integer constants, $a < b$, and $a > 1$. If $y = f(x)$ and $y = g(x)$ are graphed in the *xy*-plane, which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = a(0.63)^{-bx}$

II. $g(x) = a(1.37)^{x+2} + b$

A.

I only

B.

II only

C.

I and II

D.

Neither I nor II

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$$y = 3\left(\frac{a}{6}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 6$ and $b > c$?

A.

Zero

B.

One

C.

Two

D.

Three

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23 Mark for Review

$$\sqrt[3]{304n}(\sqrt[4]{304n})^2$$

For what value of x is the given expression equivalent to $(304n)^{12x}$, where $n > 1$?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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5 Mark for Review

The function f is defined by $f(x) = -22^x$. The function g is a decreasing linear function. In the xy-plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m), where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

A $x > k$

B $x < h$

C $x > k$ or $x < h$

D $h < x < k$

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1 Mark for Review

For the function f, for each increase in the value of x by c, where c is a positive constant, the value of $f(x)$ increases by a factor of 27. Which of the following equivalent forms of the function f displays $\frac{1}{c}$ as a coefficient of x?

A $f(x) = 35(3)^{\frac{1}{2}x}$

B $f(x) = 35(3^3)^{\frac{1}{6}x}$

C $f(x) = 35(9)^{\frac{1}{4}x}$

D $f(x) = 35(27^{\frac{1}{3}x})^{\frac{1}{2}}$

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At different times on a certain day, a manager at a theme park recorded the number of visitors. An exponential model gives the estimated number of visitors, V, at the theme park t hours after the theme park opened, where $t \leq 3$. Based on the model, 80 visitors were estimated to be at the theme park when it opened, and every 30 minutes the estimated number of visitors increased by 150% of the estimated number of visitors 30 minutes earlier. Which of the following equations best represents this model?

A) $V = 80(2.50)^{2t}$

B) $V = 80(2.50)^{\frac{t}{30}}$

C) $V = 80(1.50)^{2t}$

D) $V = 80(1.50)^{\frac{t}{30}}$

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575 The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $y = f(x)$ passes through the points $(c, 5)$ and $(2c, 137)$, where c is a constant. Which of the following could be the value of b?

A. 7

B. 16

C. 132

D. 142

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A certain investment account offers a special interest rate for the first 6 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of $\$600$ and did not make any other deposits or withdrawals. 6 months after Bennett opened the account, the balance had increased by 0.4% of the original balance. 8 months after Bennett opened the account, the balance had increased by an additional 0.3% of the balance at the end of the first 6 months. Every 2 months after the first 8 months, the balance had increased by an additional 0.3% of the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 6$?

(A) $B(x) = 602.40(1.003)^{\frac{x}{2}-6}$

(B) $B(x) = 602.40(1.003)^{\frac{x}{2}-\frac{6}{2}}$

(C) $B(x) = 602.40(1.003)^{2x-12}$

(D) $B(x) = 602.40(1.003)^{2x-6}$

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$$g(x) = \frac{1}{11}(6(2)^x + 7)$$

The function g is defined by the given equation. For all values of x, the value of $g(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

(A) 0

(B) $\frac{13}{11}$

(C) $\frac{7}{11}$

(D) 7

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